Nitrogen Blowdown Vent Test Report (NTRS-style)

Abstract

This report documents a cold-flow blowdown test of a pressurized nitrogen tank venting to ambient through a fixed orifice. The bulk of the document covers programmatic background; the runnable parameters appear in Section 4.

1. Background and Program History

The test program objectives, historical context, and programmatic background are summarized in the following discussion. Prior campaigns established the qualification basis for the subsystem, and lessons learned from earlier development informed the instrumentation plan, the data acquisition cadence, and the review gates. Personnel coordination, facility scheduling, range safety coordination, and the configuration management process are documented in the supporting appendices and are not repeated in detail here.

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4. Test Article Configuration

Working fluid: Nitrogen.

Tank internal volume: 10.0 liters.

Tank initial pressure: 5.0 MPa (5000000 Pa).

Tank initial temperature: 293.15 K.

Vent path: a single fixed orifice with effective area CdA = 1.0e-6 m^2.

Downstream boundary: ambient air at 101325 Pa and 293.15 K.

Test duration: 10.0 seconds with a 0.01 second time step.

5. Supplementary Discussion

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